TEXORA · LECTURE SHEET

Denim Production

The world's most successful special fabric, made backwards: the warp is dyed before weaving, the weft stays white, and the value is finished in the wash. The full chain — yarn, rope or slasher indigo dyeing, sizing, twill weaving, finishing and garment washing.

FM-4103 · SPECIAL FABRIC PRODUCTION · TOPIC 6 · FABRIC MANUFACTURING SPECIALIZATION

1. What denim is

Denim is a warp-faced twill fabric — classically 3/1 twill, also 2/1 and broken twills — in which the **warp is dyed (traditionally indigo) and the weft is undyed**. Because the weave is warp-faced, the face shows mostly dyed warp (blue) and the back shows mostly white weft; abrasion and washing remove indigo progressively from the warp surface, producing the fading that defines jeans aesthetics. Traditional denim is 100% cotton in medium-coarse counts (warp roughly Ne 6–16 territory for classic weights, finer for shirting denims); modern production adds polyester-core and elastane stretch denims, slub and crosshatch effects, and coloured/ecru variants — but the production logic below is the classic chain every variant modifies.

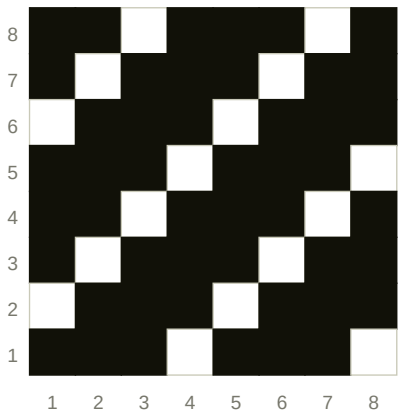


Figure 1 — The denim weave: 3/1 twill, two repeats shown (black = warp up). Each pick floats one end and binds three; the warp covers the face three-to-one, so a dyed warp over white weft gives the blue face and pale back from one dyeing of one yarn system.

2. Yarn preparation and indigo dyeing

Yarns: warp yarns are usually ring-spun for premium denim (softer, stronger, better dye behaviour) or rotor-spun for volume products (more uniform, harsher); weft yarns are softer-twisted and may be

coarser to build weight. The warp is wound as a sheet or as ropes for dyeing:

- **Rope dyeing:** the warp is formed into ropes (each of a few hundred ends) that pass repeatedly through indigo baths — dip, squeeze, then **skying** (air exposure) between dips so the reduced (leuco) indigo oxidises back to blue on the fibre. Multiple dip–sky cycles build the shade layer by layer. The ropes are then dried, long-chain beamed, and must be **re-beamed and separated** before sizing/weaving — extra handling that buys the most level, characterful dyeing.
- **Slasher (sheet) dyeing:** the warp travels as a sheet through a combined dye-and-size range: dye boxes, skying, washing, then directly into sizing and drying on the same line. Faster and cheaper, one pass, slightly less depth and character; dominant for volume denim.

The indigo mechanism (TCC chemistry from TE-2205/3105 applied): indigo is a vat dye — insoluble blue pigment reduced in alkali (traditionally with sodium hydrosulphite) to soluble yellow-green leuco form, which has affinity for cotton; on skying it oxidises back to insoluble blue *inside and on* the fibre. Indigo's peculiarity is that it deposits heavily near the fibre surface (**ring dyeing**): the core stays nearly white. Ring dyeing is not a defect — it is the product: surface dye abrades away in wear and washing, revealing white cores and the fade patterns (whiskers, honeycombs, stack fades) that the garment industry then accelerates deliberately.

3. Sizing and weaving

The dyed warp is **sized** (starch/PVA-based size, as in FM-3207's sizing calculations) to survive weaving abrasion — in slasher dyeing this happens in-line; rope-dyed warp is sized after re-beaming. **Weaving** is on air-jet or rapier/projectile looms at high speed, in 3/1 (or 2/1) twill with a warp-dominant sett: the face must be warp-covered, so ends/cm greatly exceed the value a balanced cloth of the same yarns would carry.

> **WORKED EXAMPLE — WEAVING ACCOUNT OF A DENIM**
>
> Construction: finished width 150 cm, sett 28 ends/cm, picks 25/cm, on an air-jet loom at 900 picks/min.
>
> Total warp ends = 150 × 28 = **4200 ends** — the number the warping, dyeing and beaming plan must supply (plus selvedge ends).
>
> Theoretical production = 900 × 60 ÷ (25 × 100) = 54,000 ÷ 2,500 = **21.6 m of cloth per hour** at 100% efficiency; at a realistic 88% the loom delivers about **19.0 m/h** — the figure from which loom counts for an order are planned (FM-3207 method).

4. Fabric finishing

Greige denim is finished by **singeing** (surface fibres burnt off for a clean face), **desizing** where required, **skew control** (the twill's inherent torque twists garment legs — the "leg twist" of jeans — so the cloth is deliberately skewed in finishing to compensate), **sanforizing** (compressive shrinkage to fix residual shrinkage to specification), and optionally **mercerising, coating or resin** treatments for premium surfaces. The cloth sold as "rigid" denim is then garment-washed lightly or not at all; most fashion denim is sold ready for garment processing.

5. Garment washing and effects — where the value is finished

- **Rinse/desize wash:** removes size, softens; minimal fade.
- **Stone wash:** pumice stones abrade the surface in the wash wheel — mechanical indigo removal, soft handle; stone dust and sludge are the environmental cost.
- **Enzyme wash:** cellulase enzymes hydrolyse surface cotton fibres, releasing indigo with less damage and no stones (and combined stone + enzyme grades between the extremes).
- **Bleach wash:** hypochlorite or permanganate bleaching for light and white effects; potassium permanganate sprayed locally (with neutralisation) for targeted fading.
- **Hand and machine local effects:** hand scraping/sanding, whiskering (creasing patterns set by resin and pressing), grinding at hems and pockets.
- **Laser and ozone (the modern routes):** laser fading writes fade patterns photon by photon with no water or stones and perfect repeatability; ozone gas fading bleaches in a dry process. Both answer the water, sludge and worker-exposure (sandblasting was banned in many jurisdictions for silicosis) problems of the classic routes — the clearest current direction of denim technology (compare Topic 4).
- **After-wash:** softening, fixing of shade, and quality control of shrinkage and skew in the garment state.

6. Quality and environmental notes

Denim quality is judged on shade consistency (the dyer's eternal battle with indigo), warp streaks and weft bars, shrinkage and skew after wash, colourfastness to rubbing (indigo's ring dyeing makes dry/wet crocking inherently modest — specified, not wished away) and tear/tensile retention after washing processes. Environmentally the chain's hotspots are indigo bath chemistry, wash water volume, pumice sludge and permanganate — hence the industry's move to foam and low-liquor dyeing, laser/ozone finishing and recycled-cotton yarns: denim is where textile sustainability arguments are most publicly tested.

Glossary

Rope dyeing — warp dyed as ropes through repeated dip–sky cycles. **Slasher dyeing** — sheet-form warp dyeing combined with sizing in one range. **Skying** — air oxidation of leuco indigo back to blue between dips. **Ring dyeing** — dye concentrated at the fibre surface, core near-white; the root of denim fading. **Sanforizing** — controlled compressive shrinkage finishing. **Crocking** — colour transfer by rubbing.

Key points

• Denim = warp-faced twill + dyed warp/white weft; one dyeing colours the face through the weave geometry itself.
• Indigo is applied reduced and oxidised in the air (dip–sky cycles, rope or slasher); its ring dyeing is the engineered basis of fading.
• Weaving account: 4200 ends and 21.6 m/h theoretical (≈19.0 m/h at 88%) for the worked construction.
• Finishing fixes skew and shrinkage; garment washing (stone, enzyme, bleach — increasingly laser and ozone) manufactures the final appearance.

Practice

1. Why does denim fade where it is abraded but a piece-dyed twill of the same construction does not fade the same way? Answer through dye location.
2. A denim is woven at 1100 picks/min with 22 picks/cm. Compute theoretical production per hour and the output in an 8-hour shift at 90% efficiency.
3. State the process difference between rope and slasher dyeing and one quality consequence of each.
4. Explain leg twist in jeans and the finishing operation that compensates for it.

Answer key

1. Indigo is **ring-dyed**: the colour sits at the fibre surface with a near-white core, so removing a little surface removes the colour. A piece-dyed (or fully penetrant yarn-dyed) fabric has dye distributed through the fibre cross-section, so abrasion exposes fibre that is still dyed — it wears lighter only as fast as it wears away entirely.
2. Production = 1100 × 60 ÷ (22 × 100) = 66,000 ÷ 2,200 = **30.0 m/h**; per 8 h shift at 90%: 30.0 × 8 × 0.90 = **216 m**.
3. **Rope dyeing** dyes the warp as ropes with repeated dip–sky oxidation cycles, then requires re-beaming/separation before sizing: deepest, most level and most characterful shade, at extra handling cost. **Slasher dyeing** dyes the warp as a sheet and sizes it in the same continuous range: faster and cheaper, one-pass economy, but typically slightly less depth of shade and ring-dye character.

4. The unbalanced 3/1 twill exerts a torque: after washing relaxes the cloth, the fabric skews and the garment's legs rotate ("leg twist"). Finishing compensates by deliberately **pre-skewing the cloth in the opposite sense** (weft skewing during sanforizing/finishing) by the amount the construction is known to relax, so garment washing brings the legs back to straight.
